UCT41042 – Practical for Introduction to Smart System

Department of Information & Communication Technology

Faculty of Technology

South Eastern University of Sri Lanka

Lab Sheet No: 09

Date: 01.06.2026

Title: ESP32 IoT System with MQTT and Node-RED Integration

Aims:

To design, simulate, and deploy an ESP32-based IoT system that publishes sensor data via MQTT and visualizes it using Node-RED dashboards. Learners will develop firmware, configure an MQTT broker, build Node-RED flows, and create real-time monitoring dashboards for a Smart Room Environmental Monitor.

Tasks: Smart Room Environmental Monitor

Part 1: Environment Setup

1. Install extensions in VS Code
    a. MQTT Explorer – standalone MQTT client for debugging
    b. PlatformIO IDE by PlatformIO – compile ESP32 firmware
    c. Node.js (LTS) – required for Node-RED runtime

        node –version

    d. Wokwi by Wokwi – browser-based ESP32 simulator
2. Install Node-RED globally via npm:

    npm install -g --unsafe-perm node-red

3. Install an MQTT Broker (Mosquitto):
    a. Download from https://mosquitto.org/download/
    b. After install, start the broker:
        i. mosquitto -v

c. Create file named **mosquitto.conf** and save on working folder.

    i. Type below codes:

        listener 1883

        allow_anonymous true

d. Start the broker.

    i. In cmd : *mosquitto -c mosquitto.conf -v*

Part 2: Project Setup in Wokwi / VS Code

1. Name the project **ESP32NodeRed-Lab**.
2. Circuit Design
    a. Connect the DHT22 sensor to an ESP32 GPIO pin.
    b. Connect the PIR sensor to an ESP32 GPIO pin.
    c. Connect the LED to the ESP32.
3. Library Integration
    a. PubSubClient
    b. DHT sensor library
4. Configure Node-RED
    a. Launch Node-RED
    b. Install Required Nodes
        i. @flowfuse/node-red-dashboard

c. Build the MQTT Subscriber Flow

Node	Type	Configuration
Temperature Flow		
`mqtt in`	Input	Topic: home/room/temperature, Server: ip_address:1883
`gauge`	Dashboard	Label: Temperature, Min: 0, Max: 80, Unit: °C
`chart`	Dashboard	Label: Temperature History
Humidity Flow		
`mqtt in`	Input	Topic: home/room/humidity Server: ip_address:1883
`gauge`	Dashboard	Label: Humidity, Min: 0, Max: 100, Unit: %
Motion Detection Flow		
`mqtt in`	Input	Topic: home/room/motion
`text`	Dashboard	Label: Motion Status
LED Control Flow		
`switch`	Dashboard	Label: LED Control, On Payload: ON, Off Payload: OFF
`mqtt out`	Output	Topic: home/room/led/set, Server: ip_address:1883

d. Click Deploy (top-right red button) after wiring all nodes.

e. Open the dashboard at http://localhost:1880/dashboard to view live data.

5. Programming Logic Development in Wokwi / VS Code
    a. Initialize Wi-Fi and MQTT connection to enable network communication between ESP32 and the broker.
    b. Read temperature and humidity values from the DHT22 sensor using the appropriate sensor library.
    c. Read and process PIR sensor data for motion detection.
    d. Publish all sensor readings to MQTT topics in real time for Node-RED dashboard visualization.

6. Simulation & Testing in Wokwi / VS Code
    a. Run simulation
    b. Adjust values in Wokwi / VS Code
    c. Verify Node-RED interaction